

# Pharmacist Telemonitoring of Antidepressant Use: Effects on Pharmacist–Patient Collaboration

Nathaniel M. Rickles, Bonnie L. Svarstad, Jamie L. Statz-Paynter, Leslie V. Taylor, and Kenneth A. Kobak

## ABSTRACT

**Objective:** To explore the impact of telephone-based education and monitoring by community pharmacists on multiple outcomes of pharmacist–patient collaboration.

**Design:** A randomized, controlled, unblinded, mixed experimental design.

**Setting:** Eight Wisconsin community pharmacies within a large managed care organization.

**Patients:** A total of 63 patients presenting new antidepressant prescriptions to their community pharmacies.

**Interventions:** Patients were randomized to receive either three monthly telephone calls from pharmacists providing pharmacist-guided education and monitoring (PGEM) or usual pharmacist's care. Usual care is defined as that education and monitoring which pharmacists may typically provide patients at the study pharmacies.

**Main Outcome Measures:** Patient's frequency of feedback with the pharmacist, antidepressant knowledge, antidepressant beliefs, antidepressant adherence at 3 and 6 months, improvement in depression symptoms, and orientation toward treatment progress.

**Results:** Of the 60 patients who completed the study, 28 received PGEM and 32 received usual pharmacist's care. Results showed that PGEM had a significant and positive effect on patient feedback, knowledge, medication beliefs, and perceptions of progress. There were no significant group differences in patient adherence or symptoms at 3 months; however, PGEM patients who completed the protocol missed fewer doses than did the usual care group at 6 months ( $P \leq .05$ ).

**Conclusion:** Antidepressant telemonitoring by community pharmacists can significantly and positively affect patient feedback and collaboration with pharmacists. Longer-term studies with larger samples are needed to assess the generalizability of findings. Future research also needs to explore additional ways to improve clinical outcomes.

**Keywords:** Antidepressant, community and ambulatory pharmacy, telephone monitoring, counseling, adherence, outcomes.

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Depression is a serious and common mental disorder with a lifetime risk of 10% to 25% of women and 5% to 12% of men.<sup>1</sup> While antidepressant medications are relatively effective for depression symptoms, a high rate of nonadherence occurs with these medications, and more than one third of patients discontinue their medication within the first 3 months of treatment.<sup>2</sup> Premature discontinuation contributes to a high relapse rate and poor treatment outcomes.<sup>3–5</sup>

Inadequate provider–patient communication is believed to play a major role in the development of poor antidepressant outcomes.<sup>2,6–8</sup> In a national study, fewer than 20% of the 1,001 participating patients indicated having been told about common antidepressant adverse effects such as sexual problems and insomnia.<sup>6</sup> In another study, only 34% of patients indicated that they were told about the need to continue their antidepressant for at least 6 months, and almost one half of those who experienced adverse effects did not discuss them with their physician or pharmacist.<sup>7</sup> As might be expected, patients who discussed treatment duration and/or medication concerns with their providers were more likely to adhere to their antidepressant regimens than those who did not have such discussions.<sup>7,8</sup> Studies also indicate that case managers, nurses, and master’s-level therapists can improve

outcomes by calling patients to provide educational reinforcement and to discuss antidepressant adverse effects, treatment responses, and adherence.<sup>9–12</sup>

Several studies indicate that pharmacist monitoring of other psychiatric medications can reduce adverse effects and rehospitalizations.<sup>13–15</sup> However, only two published studies have reported on pharmacists’ role in antidepressant education and monitoring.<sup>16,17</sup> The first study involved a prospective evaluation of the impact of clinic-based pharmacists who provided a combination of services for new antidepressant users, including office visits, telephone calls, e-mail communications to prescribers with recommendations and issues, and weekly pharmacist–prescriber discussions.<sup>16</sup> The interventions had no impact on patient adherence at 3 months, but significant improvement in adherence at 6 months, greater medication switches and satisfaction with care at 6 months, and fewer visits to prescribers at 12 months after study enrollment were observed. One limitation of this study was that it required office visits and was labor intensive, making it difficult to implement in community pharmacy practice. A second limitation was that it emphasized traditional biomedical roles and did not determine whether the intervention actually improved patient knowledge, antidepressant beliefs, patient feedback regarding adverse effects, or patient perceptions of treatment progress.

The second study explored the role of community pharmacists in antidepressant education and monitoring.<sup>17</sup> In a prospective field study of 100 antidepressant users, Bultman and Svarstad<sup>17</sup> found that antidepressant monitoring by community pharmacists was significantly associated with greater patient satisfaction with antidepressants and adherence. Unfortunately, the effects of antidepressant monitoring on patient feedback or patient orientation toward treatment progress were not assessed in this study. Another limitation of this study was that it did not use an experimental design, making it difficult to test the relationships hypothesized in Svarstad and Bultman’s Health Collaboration Model (HCM).<sup>18</sup> In the HCM, quality medication education and collaborative monitoring by health care providers is posited to result in improved cognitive, behavioral, and clinical outcomes, including better patient comprehension and recall of the regimen, more positive beliefs about the medication and its effects, more frequent patient feedback regarding adherence and adverse effects, greater adherence to the prescribed regimen, and better clinical outcomes.

Several well-controlled studies across multiple disease states have shown that community pharmacists can have a significant impact on patient knowledge about their illness and treatment,<sup>19</sup> medication adherence,<sup>20–22</sup> and clinical outcomes.<sup>19,21–24</sup> Two studies<sup>21,22</sup> were also important in showing that educational and monitoring studies could be both brief and effective. However, we found no experimental studies exploring these specific relationships and findings among antidepressant users.

## AT A GLANCE

**Synopsis:** Feedback to community pharmacists was significantly greater from patients who received pharmacist-guided education and monitoring (PGEM) for antidepressant use than from those who received pharmacist’s usual care. To assess a relatively simple medication monitoring tool—one 90-minute pharmacist training session and three pharmacist–patient telephone calls—a total of 60 patients in eight community pharmacies were monitored during their first 3 months of antidepressant use. Patients who received telemonitoring demonstrated significantly better knowledge, medication beliefs, and perceptions of progress than did patients who received usual care. The feedback provided to pharmacists by PGEM patients increased collaboration and enabled pharmacists to provide education and guidance tailored to individual patient needs.

**Analysis:** *Although antidepressant medications are relatively effective for relief of depression symptoms, a high rate of nonadherence and nonpersistence occurs with these medications, and more than one third of patients discontinue use within the first 3 months of therapy. The role of community pharmacists in increasing patient adherence and persistence with antidepressant therapy has not been explored by using brief intervention techniques in an experimental design. A simple pharmacist intervention can have significant positive effects on several outcomes measures.*

## Objective

The objective of this study was to explore the impact of telephone-based, pharmacist-guided education and monitoring (PGEM) on multiple outcomes of pharmacist–patient collaboration and monitoring in eight community pharmacies. Building on past research and the HCM, we hypothesized that, when compared with usual care, collaborative antidepressant education and monitoring by community pharmacists would yield (1) greater frequency of patient feedback to pharmacist, (2) fewer missed antidepressant doses, (3) greater antidepressant knowledge, (4) more positive antidepressant beliefs, (5) a more positive orientation toward treatment progress, and (6) greater improvement in depression symptoms.

## Methods

Study procedures are summarized in Figure 1. The University of Wisconsin–Madison Health Sciences Human Subject Committee approved these procedures in July 2001. Enrollment began on October 1, 2001, and ended on September 30, 2002.

### Pharmacist Recruitment

A total of 14 pharmacists were recruited from eight Wisconsin community pharmacies affiliated with a large managed health care organization. Pharmacists were eligible to participate if they were fully licensed, worked full-time for one of the pharmacies, and consented to participate in a paid (\$100) 90-minute training session and other study procedures. The training session discussed how to recruit patients and how to use a two-page monitoring tool developed for pharmacists' use during three monthly telephone calls to patients in the experimental group. The monitoring tool guided pharmacists to ask patients specific questions about their antidepressant knowledge, beliefs, adverse effects, and other concerns, adherence, and progress toward patient-identified treatment goals. The tool also guided pharmacists on how to respond to different issues and concerns and how to document their interventions. They were instructed to provide the usual care to those patients assigned to the usual care group (i.e., no special counseling, monitoring of adherence, or telephone follow-up). Usual care was defined as the education and monitoring patients typically received at a study pharmacy.

### Patient Recruitment

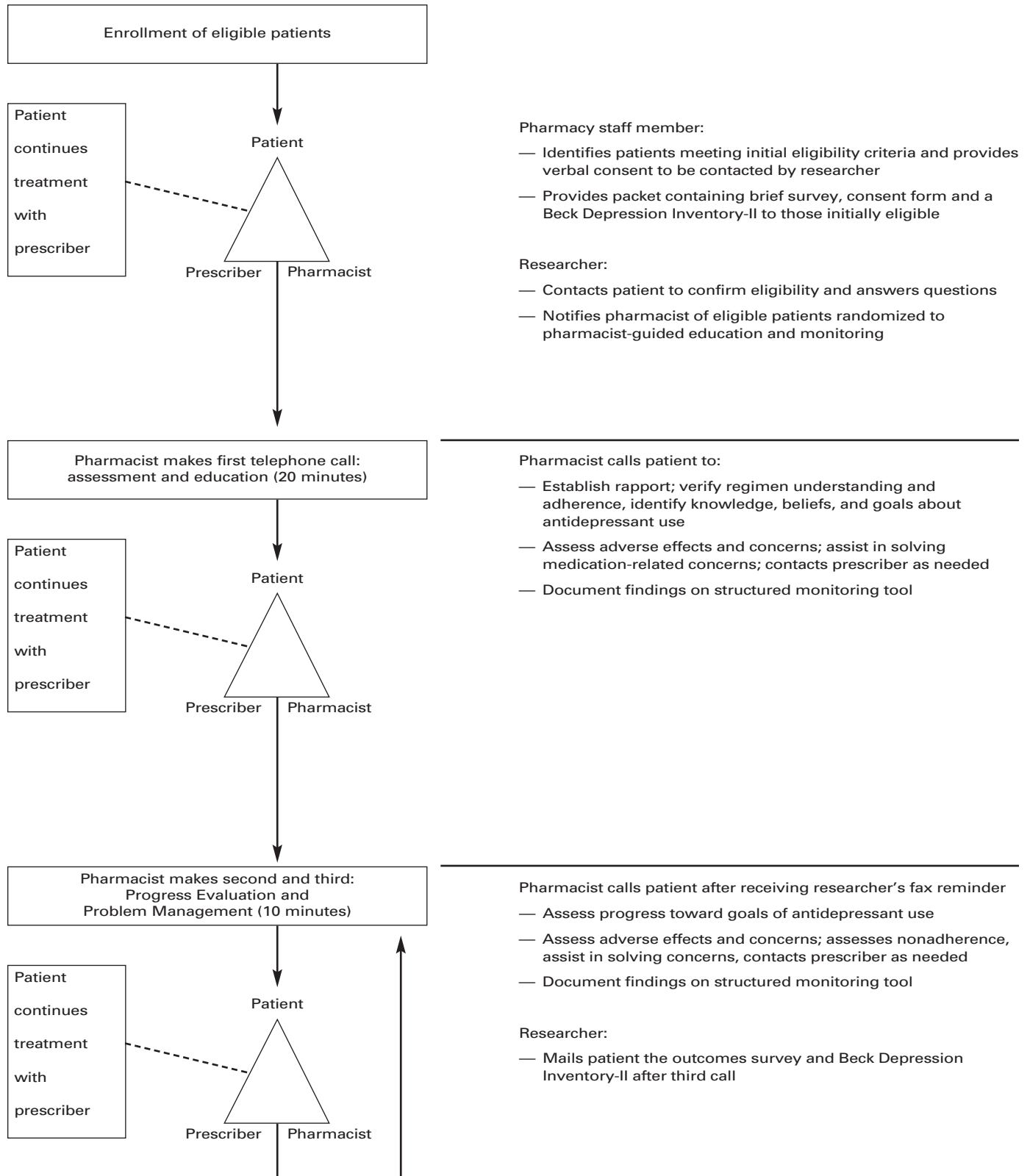
Pharmacy staff identified potentially eligible patients by identifying antidepressant prescriptions and checking patient profiles to determine whether the patients had received the medication within the past 4 months. Once identified, these patients were given a one-page information sheet to read and complete if they were interested in the study. This information sheet informed patients that some of them would receive a new medication monitoring ser-

vice involving brief telephone calls and medication assistance from a pharmacist while others would continue to receive the follow-up care they typically receive at the pharmacy. The information sheet was also used to screen patients for eligibility. Patients were eligible if they had no antidepressant use in the past 4 months,<sup>10</sup> were 18 years or older, were willing to pick up their antidepressant from a study pharmacy during the next 4 months, had no hearing impairment, and planned to be in the local area during the next 4 months. Patients who were eligible were instructed they would be contacted by the researcher and were given a study packet containing a consent form and Beck Depression Inventory–II (BDI-II).<sup>25</sup> The consent form requested permission for access to prescription refill records only and informed patients they would receive \$15 for participation. The researcher confirmed eligibility and asked patients to return the completed consent form and baseline BDI-II promptly. Patients were excluded if they had a BDI-II score below 16, required a translator, were pregnant or nursing, were receiving medications for a psychotic or bipolar disorder, and/or had physical conditions requiring additional caution with their antidepressant. A BDI-II score of 16 was chosen as a cutoff since it is a midpoint in the published range of 14–19 that indicates mild depression.<sup>25</sup> Sample size calculations were based on past research and Cohen's power tables.<sup>26,27</sup> A meta-analysis was conducted showing the impact of educational and behavioral interventions on adherence measures using refill records.<sup>26</sup> This meta-analysis showed a mean effect size of 0.73 for such interventions. At a conventional power of 0.80, alpha of 0.05 and an effect size of 0.70, power calculations resulted in a sample size requirement of at least 26 patients per each study group (a total of 52 patients).<sup>27</sup>

Randomization involved the researcher preparing 10 pieces of paper with sequential numbers for each participating pharmacist at the site (i.e., sites with two pharmacists would have 20 potential numbers, three pharmacists at a site would have 30 numbers). Each of the eight pharmacies had a different cluster of numbers; the first site had numbers beginning with 100, the second site had numbers with 200, and so forth. Since the first site had three participating study pharmacists, 30 slips of paper numbered from 100 to 130 were prepared and placed into an envelope. When a patient was enrolled from that site, the researcher would randomly select a number out of the envelope. Selection of an odd or even number meant the patient was assigned to the usual care or PGEM group, respectively. Patients were unblinded to their group assignment since they could figure out that if they did not receive telephone calls, they were likely assigned to the usual care group. No "dummy" telephone calls (calls that did not involve the content of the intervention) were made to the usual care group to simulate the intervention.

### Pharmacist-Guided Education and Monitoring

Patients randomized to the PGEM group received 3 monthly telephone calls from a study pharmacist. The decision to focus on

**Figure 1. Pharmacist-Guided Education and Monitoring and Process of Care**

the first 3 months of treatment stemmed from previous research<sup>28</sup> that described this period of time as the “acute phase of treatment” associated with a high rate of antidepressant nonadherence.<sup>2</sup> Therefore, we believed PGEM might have its greatest impact during the first 3 months of antidepressant use. The first telephone call took place, on average, within the first 3 weeks of the patient picking up their initial antidepressant prescription from the pharmacy. During the first call, the pharmacist assessed the patient’s antidepressant knowledge and beliefs, adverse effects and other concerns, treatment goals or areas in which they hoped the medication would help, and how the medication was being used during the week before the telephone call. Study pharmacists probed and clarified or explained issues that were not understood by patients. They also asked patients to rate the severity of their concerns and made recommendations on how to handle any adverse effects, difficulties remembering or paying for medications, and other concerns. Pharmacists were expected to follow up on any indication of medication nonadherence using supportive probing to inquire why the doses were missed and making recommendations to help the patients better use the medication. On average, the first telephone call took study pharmacists 19 minutes to complete.

The second and third telephone calls took place approximately 1 and 2 months after the initial call. In a majority of cases, the pharmacist who made the first telephone call to the patient made the second and third telephone calls. To enhance timely follow-up, the research pharmacist sent fax reminders to study pharmacists when patients were due to receive a telephone call. During these patient calls, study pharmacists used the monitoring tool to guide their follow-up on any issues or concerns identified in earlier calls. Pharmacists also reviewed current adherence, whether any new adverse effects and concerns had developed, and whether the patient had seen any progress in his/her medication goals. The pharmacist made new recommendations to patients as needed. On average, the second and third telephone calls required 12 and 11 minutes, respectively, of pharmacists’ time. In the initial training session, study pharmacists were encouraged to contact prescribers if concerns or questions arose. However, no reminders or tools were provided to enhance or guide pharmacist communication with prescribers.

## Measures of Behavioral Outcomes

Two types of behavioral outcomes were examined in this study. The first measure included seven items in the outcomes survey (given at the end of each patient’s 3-month study period) examining the frequency of patient feedback to pharmacist (FPFP) during the study period. This measure was intended to capture patient feedback via various modes such as providing feedback to the pharmacists in response to the pharmacist-initiated telephone calls (such as in the PGEM group), talking with pharmacists in the pharmacy, or calling the pharmacists themselves. The items were developed for this study and asked patients to estimate how frequently they told the pharmacist (1) what they knew about the antidepressant, (2) changes in symptoms, (3) if their aims of

antidepressant therapy were being met, (4) adverse effects, (5) other concerns with the antidepressant, (6) feelings they had about using the medication, and (7) how they were using the antidepressant. Response categories included: never, 1; rarely, 2; sometimes, 3; frequently, 4; and “does not apply.” Patients who checked “does not apply” or none of the other categories were counted as “never.” Factor analysis indicated that all items loaded on one factor and explained 72% of the variance. The scale had a Cronbach alpha of 0.93, showing excellent internal consistency.<sup>29</sup> Responses were summed across items (possible range, 7 to 28) and then averaged by dividing the total sum score by the number of items answered.

The second type of behavioral measure, medication adherence, was constructed from pharmacy records over two time periods (the first 3 months after enrollment and the second 3 months after enrollment). We measured adherence in both time periods, because past research suggests that adherence declines over time and that providers may need more than 3 months to resolve such problems.<sup>16</sup> In the present study, the number of missed doses was calculated by multiplying the number of prescribed doses per day times the number of days late between refills for the first 3-month period and second 3-month period. Results were multiplied by 100 to yield the percentage of missed doses for each period.

Since adherence is difficult to measure objectively,<sup>30</sup> we validated the pharmacy records in two ways. First, we compared pharmacy records with prescription insurance claims for 49 of 63 patients for whom claims data were available from the participating managed care organization (3 patients paid cash for their medications, and 11 others had insurance outside the managed care organization). The available pharmacy records and claims data were fairly consistent. Inconsistencies were resolved by the research pharmacist after case-by-case analysis. Second, we measured the patient’s self-reported antidepressant adherence as part of the outcomes survey. Patients were asked “in the past 7 days ending yesterday, how many times did you miss taking a pill?” This item was similar to an item used in the Brief Medication Questionnaire developed and validated by Svarstad and colleagues.<sup>31</sup> They found that a well-designed, 1-week self-report adherence measure can have excellent predictive validity when compared with the true rate of dose omission as measured by Medication Events Monitoring Systems (MEMS) over a 7-day or 30-day period.<sup>31</sup> In the present study, we found a significant correlation between the 7-day self-reported missed doses and missed doses according to pharmacy records over the first 3-month period ( $r = 0.760$ ;  $P \leq .001$ ). Since pharmacy record data were available for all patients over a longer period, they are used in this report.

## Measures of Cognitive Outcomes

The outcomes survey evaluated three cognitive outcomes: antidepressant knowledge, beliefs, and orientation toward treatment progress (OTTP). Antidepressant knowledge was measured by three items adapted from previously published measures.<sup>17</sup> A close-ended format was used to minimize respondent burden and

the need for researcher interpretation. The items assessed knowledge of (1) how the antidepressant worked, (2) how long individuals who receive antidepressants for the first time should be on the medications, and (3) how long it takes for antidepressants to start working. Each item was scored as correct (1) or incorrect (0). Responses were summed across items (possible range, 0 to 3) and then averaged by dividing the total sum score by the number of items answered.

Four antidepressant belief items were developed based on prior research and the Health Belief Model.<sup>17,32</sup> Using a Likert scale from strongly disagree, 1, to strongly agree, 5, patients were asked to rate their agreement with beliefs that (1) depression requires the use of medications, (2) antidepressants are effective for the patient's symptoms, (3) antidepressants can cause physical dependence, and (4) antidepressants can cause long-term harm to the body. Factor analysis revealed one construct underlying this scale that explained 56.1% of the variance. The Cronbach alpha was 0.72, indicating acceptable internal consistency.<sup>29</sup> Responses were summed across items (possible range, 4 to 20) and then averaged by dividing the total sum score by the number of items answered.

The third cognitive outcome measured the patient's awareness of their treatment progress or OTTP. Three items asked patients to rate their agreement (strongly agree, 1; strongly disagree, 5) that conversations with pharmacists during the previous 3 months helped them become more aware and better able to describe and measure how the antidepressant helped them achieve their goals. Factor analysis indicated that the scale had one underlying factor that explained 90.9% of the variance. The Cronbach alpha was 0.95, indicating excellent internal consistency.<sup>29</sup> Responses were summed across items (possible range, 3 to 15) and then averaged by dividing the total sum score by the number of items answered.

## Measures of Clinical Outcomes

The BDI-II was used to measure depression symptoms.<sup>25</sup> The BDI-II is a self-report instrument with acceptable psychometric properties and brief administration time.<sup>25</sup> Patients rated the severity of 21 symptoms using four response categories scored from 0 (no symptoms) to 3 (most severe symptoms). Beck and colleagues suggest that total scores of 0–13 indicate minimal depression, 14–19 mild depression, 20–28 moderate depression, and 29–63 severe depression.<sup>25</sup> We examined mean BDI-II scores and a dichotomous variable that measured whether the patient experienced at least a 50% reduction in BDI-II scores from the beginning to the end of the 3-month intervention period. Other researchers have used similar dichotomous measures to assess improvement in depression symptoms.<sup>9,11,33</sup>

## Data Analysis

After computing descriptive statistics, *t* tests were used to compare differences between the PGEM and usual care groups on antidepressant knowledge, antidepressant beliefs, OTTP, FFPF,

and adherence. Chi-square tests were used to compare differences between the PGEM and usual care groups on whether patients experienced 50% or more improvement in BDI-II scores. The Fisher exact test was used for testing group differences involving categorical data with less than 5 cases per cell.<sup>34</sup> One-tailed tests were used to test unidirectional hypotheses. The *P* level was set a priori at .05.

## Results

### Patient Characteristics

A total of 98 patients indicated initial interest in the study. Of these, 35 were defined as ineligible because they did not return the consent form and baseline materials (*n* = 10), had a baseline BDI-II score below 16 (*n* = 14), could not be reached by the researcher (*n* = 4), used antidepressants in the past 4 months (*n* = 2), or decided not to participate (*n* = 5). The remaining 63 patients who consented and scored a 16 or higher on the BDI-II were randomized by the researchers to receive PGEM (*n* = 31) or usual pharmacist's care (*n* = 32). Of 31 PGEM patients, 28 completed the study protocol and 3 patients were lost to contact before the third telephone call. These 3 dropouts came from different pharmacies and pharmacists.

The data in Table 1 describe the characteristics of 63 patients who were enrolled in the study. Of these patients, 92% were white and 84% were women. On average, patients were 38 years old (range, 19 to 70), had greater than high school education, and took no other medications. The mean baseline BDI-II score was 28, indicating moderate depression. Most patients (94%) received no telephone calls from a pharmacist in the 3 months prior to the study. The two study groups did not vary significantly by gender, age, race, education, other medications, or previous calls from pharmacists. Despite randomization, PGEM patients were more likely than usual care patients to have a history of psychotropic medication use (*P* ≤ .05).

### Impact of PGEM on Behavioral Outcomes

As shown in Table 2, the mean of the total FFPF scale was significantly higher in the PGEM group than in the usual care group (mean, 23 versus 11, *P* ≤ .001). PGEM patients were significantly more likely than usual care patients to give feedback to pharmacists regarding their knowledge, changes in symptoms, whether their aims were being met, adverse effects, other antidepressant issues, and how they were using the antidepressant (*P* ≤ .001). In short, PGEM patients were more likely than other patients to communicate their antidepressant experiences to study pharmacists.

Data in Table 2 also show the effects of PGEM on antidepressant adherence among the 60 patients who remained in the study for the 3-month intervention period. While there was no significant difference between study groups over the first 3 months, the rate of missed doses at 6 months was significantly lower in the PGEM group than the usual care group (30% versus 49%, *P* ≤ .05). When

**Table 1. Patient Characteristics by Study Groups (n = 63)**

| Characteristic                                                             | PGEM<br>(n = 31) | Usual Care<br>(n = 32) |
|----------------------------------------------------------------------------|------------------|------------------------|
| <i>Gender</i>                                                              |                  |                        |
| Men, No. (%)                                                               | 6 (19.4)         | 4 (12.5)               |
| Women, No. (%)                                                             | 25 (80.6)        | 28 (87.5)              |
| <i>Age in years, mean ± SD</i>                                             | 37.8 ± 10.7      | 37.5 ± 13.4            |
| <i>Race</i>                                                                |                  |                        |
| White, No. (%)                                                             | 27 (87.1)        | 31 (96.9)              |
| Other, No. (%)                                                             | 4 (12.9)         | 1 (3.1)                |
| <i>Education</i>                                                           |                  |                        |
| High school or less, No. (%)                                               | 8 (25.8)         | 8 (25.0)               |
| Any post-high school education,<br>No. (%)                                 | 23 (74.2)        | 24 (75.0)              |
| <i>Medication use</i>                                                      |                  |                        |
| Current number of medications<br>other than antidepressants,<br>mean ± SD  | 0.87 ± 1.41      | 0.78 ± 1.16            |
| No past history of psychiatric<br>medication use, No. (%)                  | 18 (58.1)        | 27 (84.4)              |
| Past use of psychiatric<br>medications, No. (%)                            | 13 (41.9)        | 5 (15.6) <sup>a</sup>  |
| <i>BDI-II (baseline), mean ± SD</i>                                        | 28.9 ± 8.15      | 27.0 ± 8.40            |
| <i>Telephone calls from pharmacist<br/>in past 3 months prior to study</i> |                  |                        |
| No telephone call, No. (%)                                                 | 30 (96.8)        | 29 (90.6)              |
| One or more telephone calls,<br>No. (%)                                    | 1 (3.2)          | 3 (9.4)                |

Abbreviations used: BDI-II, Beck Depression Inventory–II; PGEM, pharmacist-guided education and monitoring.

<sup>a</sup>  $P \leq .05$ .

we performed an intention-to-treat analysis that includes the three patient dropouts, this 6-month difference was not significant (data not shown). We explored the possibility that self-reported adherence varied by study group, but no significant difference was found (data not shown).

### Impact of PGEM on Cognitive Outcomes

PGEM had a significant impact on three cognitive outcomes: antidepressant knowledge, antidepressant beliefs, and OTTP. The data in Table 2 indicate the PGEM group had a significantly higher overall knowledge score than did the usual care group ( $P \leq .05$ ). Item analyses indicated that most patients could explain how antidepressants work and the onset of antidepressant effects, but only 61% of all patients knew or gave a correct answer when asked how long the antidepressant should be used. PGEM patients were more likely to respond correctly than usual care patients when

asked about the recommended duration of use (75% versus 48%,  $P \leq .05$ ). Data in Table 2 also indicate that PGEM patients had a significantly higher overall belief score than did usual care patients ( $P \leq .01$ ). Item analyses revealed significant differences between study groups on two of four beliefs; a greater number of PGEM patients than usual care patients disagreed that one can become physically dependent on antidepressants over time ( $P \leq .001$ ) and that daily use of antidepressants can be harmful ( $P \leq .05$ ).

The data in Table 2 indicate that PGEM patients had a significantly more positive orientation toward treatment progress than did the usual care patients ( $P \leq .001$ ). Item analyses indicated that the PGEM group had a significantly higher score than the usual care group across all three items of the OTTP scale. That is, PGEM significantly improved the patient's awareness of how the antidepressant helped reach personal treatment goals ( $P \leq .001$ ), the patient's ability to describe these benefits ( $P \leq .001$ ), and the patient's ability to measure progress in terms of what he or she had hoped to experience and how he or she currently felt ( $P \leq .001$ ).

### Impact of PGEM on Clinical Outcomes

An examination of BDI-II mean scores and the dichotomous measure of improvement showed that PGEM did not have a significant impact on depression symptoms. Patients in both groups experienced similar reductions in symptoms from baseline to the end of the 3-month intervention period (Table 2). Both groups showed significant reductions in symptoms from baseline to the end of the 3-month intervention period ( $P \leq .001$ ). Similarly, the percentages of patients experiencing at least a 50% improvement in BDI-II scores were not significantly different in the two groups.

## Discussion

Patients receiving consistent pharmacist-guided education and monitoring over the telephone during the first 3 months of antidepressant treatment experienced several significantly improved outcomes, compared with patients receiving the usual care from community pharmacists. Bivariate analyses confirmed four of the six study hypotheses that predicted PGEM would have a greater impact than usual pharmacist's care on the patient's knowledge, beliefs, orientation toward treatment progress, and frequency of feedback to pharmacists. Using per-protocol analyses, the hypothesis predicting PGEM's differential impact over usual care on adherence at 6 months was partially supported. Results did not support the hypothesis that predicted PGEM would have a greater impact on improving depression symptoms, compared with usual care.

This is the first known study to test whether collaborative, telephone-based education and monitoring by a pharmacist can enhance the patient's orientation toward treatment progress and frequency of feedback to pharmacists. As predicted, patients who received telephone monitoring were significantly more likely than

**Table 2. Bivariate Results of PGEM and Usual Care's Impact on Behavioral, Cognitive, and Clinical Outcomes<sup>a</sup>**

| Outcome                                    | PGEM<br>No. (Mean $\pm$ SD) | Usual Care<br>No. (Mean $\pm$ SD) | T Test             |
|--------------------------------------------|-----------------------------|-----------------------------------|--------------------|
| FPPF scale                                 | 28 (22.7 $\pm$ 4.83)        | 32 (10.9 $\pm$ 4.32)              | 101.2 <sup>d</sup> |
| Nonadherence                               |                             |                                   |                    |
| % omitted antidepressant doses at 3 months | 28 (18.1 $\pm$ 23.5)        | 32 (18.7 $\pm$ 22.1)              | 0.011              |
| % omitted antidepressant doses at 6 months | 28 (30.3 $\pm$ 36.4)        | 32 (48.6 $\pm$ 39.2)              | 3.45 <sup>b</sup>  |
| Antidepressant knowledge score             | 28 (2.54 $\pm$ 0.744)       | 31 (2.06 $\pm$ 0.929)             | 3.27 <sup>b</sup>  |
| Antidepressant belief scale                | 28 (15.7 $\pm$ 2.84)        | 31 (14.0 $\pm$ 2.32)              | 6.22 <sup>c</sup>  |
| OTTP scale                                 | 27 (12.4 $\pm$ 2.50)        | 30 (9.37 $\pm$ 3.22)              | 15.2 <sup>d</sup>  |
| Improvement in BDI-II                      | No. (%)                     | No. (%)                           | Chi-Square Test    |
| $\geq$ 50% improvement in BDI-II           | 21 (75.0)                   | 21 (65.6)                         |                    |
| < 50% improvement in BDI-II                | 7 (25.0)                    | 11 (34.4)                         | 0.625              |

Abbreviations used: BDI-II, Beck Depression Inventory–II; FPPF, frequency of patient feedback with the pharmacist; OTTP, orientation toward treatment progress; PGEM, pharmacist-guided education and monitoring.

<sup>a</sup>Results shown are based on a per-protocol analyses and excludes 3 dropouts. An intention-to-treat analysis indicates no significant differences between study groups on adherence at 3 and 6 months. Number of cases varies due to missing data.

<sup>b</sup> $P \leq .05$  (one-tailed).

<sup>c</sup> $P \leq .01$  (one-tailed).

<sup>d</sup> $P \leq .001$  (one-tailed).

usual care patients to have positive perceptions about how their medication was helping them achieve their goals and to report feedback to the pharmacists about different aspects of their medication use. Such increased patient feedback to pharmacists can be extremely valuable when tailoring therapeutic recommendations toward patient-specific needs. Further study of patient feedback and its specific role in problem-solving would be very helpful.

Our findings revealed that PGEM had a significant impact on the patient's knowledge of how long they should expect to be on the antidepressant. This is an important finding because previous work shows that almost one half of the 100 patients in the study lacked a clear understanding of the expected length of treatment<sup>8</sup> and that this lack of understanding is linked to premature discontinuation of antidepressant therapy.<sup>7</sup> In contrast to past work,<sup>8</sup> a large majority of the study patients knew how the medication worked. This discrepancy may be due to differences in the measurement of patient medication knowledge, an issue requiring further study.

The results relating to the antidepressant belief scale provide overall support for the hypothesis that PGEM has a significant impact on antidepressant beliefs. Specifically, PGEM appeared to have a significant role in reducing two negative beliefs regarding potential physical harm and dependence. There were no significant differences between the study groups regarding the perceived need for and the effectiveness of antidepressants. Study pharmacists may have had a limited impact on these beliefs since many patients may have resolved their issues about these beliefs before picking up their medications at the pharmacy. Whether similar results

would be obtained if pharmacist–patient communication began at an earlier stage in the drug use process is not known.

While PGEM had no impact on adherence at 3 months, a per-protocol analysis revealed significantly improved adherence at 6 months for those who completed the study. When the three patients who withdrew from the study were included in the analysis, the difference did not reach significance at the .05 level. The difference in results associated with the inclusion of the three dropouts is likely because the study may not have been sufficiently powered. An initial power analysis at the start of the study indicated a minimum number of 52 patients to detect significant differences in adherence across study groups. Even though we obtained a few patients beyond this minimum, the power analysis only provided an estimate. The calculations may have underestimated the sample size needed to detect a significant difference in adherence. Such insufficient power makes the significance of study findings sensitive to small changes in sample sizes. Interestingly, Finley and colleagues<sup>16</sup> found that pharmacist education and monitoring significantly improved antidepressant adherence at 6 months but not 3 months.

Antidepressant education and monitoring may be more likely to result in greater adherence at 6 months than at 3 months for two main reasons. First, patient motivation to persist with any chronic drug regimen is generally believed to decline over time, and regular professional supervision and reinforcement is needed to counteract this problem. Second, the PGEM monitoring tool clearly emphasized the importance of assessing and reinforcing the importance of taking the antidepressant for at least 4 to 6 months.<sup>28</sup>

As noted earlier, only one third of all patients report that they are given this information by their prescriber.<sup>7</sup>

No support was found for the hypothesis that PGEM would produce greater improvement in depression symptoms than would usual pharmacist's care. Both groups had significant improvements during the 3-month intervention period. This result was unexpected in light of prior research showing a differential impact of telephone-based counseling on symptoms.<sup>9–12,16</sup> Two possible explanations might explain these inconsistent outcomes. First, differences in patient selection, symptom measurement, and length of follow-up may partially account for the observed differences across studies. Second, both study groups may have been surprised when approached by community pharmacists with a special interest in mental health issues. Indeed, the invitation to participate in this study may have biased results in several ways due to the Hawthorne effect.<sup>35</sup> Both study groups may have reported greater clinical improvement since they wanted to meet the expectations of interested pharmacists (especially since patients were unblinded to group assignment). This phenomenon may explain why study groups had greater antidepressant adherence than has been reported in the literature.

## Limitations

Our study had several limitations. First, the intervention period was only 3 months in duration. Previous research has involved longer periods for antidepressant education and monitoring.<sup>11,16</sup> Educational and monitoring programs may require longer intervention periods because of the complexities and efforts associated with motivating behavioral change.

Second, the sample was relatively small and limited to one network of pharmacies in one geographic location. This limitation affects the generalizability of the results to other settings and groups of people. Larger samples in other settings are needed to further assess the value of telephone monitoring by community pharmacists.

Third, our study used several new measures that may be associated with various measurement errors. Such measurement errors can be reduced by further testing the measures for validity and reliability.

Fourth, we lacked access to prescription insurance claims for all patients. Such access would have allowed researchers to better verify whether medications had been picked up in pharmacies outside the study pharmacies and, thus, yielded a more complete picture of patient medication use. Future evaluations might consider using the MEMS, an adherence measure providing detailed and objective information about the number and timing of doses removed from the medication bottle.<sup>30,31</sup>

A fifth study limitation pertains to randomization procedures. Since we randomized by patient and not by site, patients in the usual care group may have been exposed to improved antidepressant education and monitoring when they returned for refills. Thus,

it is possible that the usual care group received enhanced care by study pharmacists. Since we found significant group differences across multiple outcomes, it is reasonable to conclude that any contamination or crossover effects were not significant or pervasive. Also, despite randomization procedures, the PGEM group had a significantly greater number of previous psychotropic medication users. This "incomplete randomization" could contribute to PGEM patients having greater knowledge and more positive beliefs than the usual care patients before the start of the intervention and subsequently contribute to the differences in cognitive outcomes across study groups at the end of the study. Finally, this study was focused on the patient's perspective and did not address how pharmacist-provider communication affected various patient outcomes. This issue should be explored in future work.

## Conclusion

Study results not only confirm previous research<sup>18</sup> describing the impact of provider-patient collaboration on behavioral and cognitive outcomes but also introduce the specific and valuable contributions of community pharmacists in antidepressant education and monitoring. Our study went beyond previous studies by using an experimental design to evaluate the role of community pharmacists in antidepressant education and monitoring. Results highlight the significant value of providing community pharmacists some basic training and structured tools with which they can help better educate and more consistently monitor new users of medication. Clearly, pharmacists can improve the nature and extent of information provided by patients to pharmacists by consistently eliciting feedback from patients on how the medication is helping them and what kinds of medication concerns they have. Pharmacists should find such additional information very beneficial as they actively collaborate with patients on ways to maximize their medication experiences.

More research is needed to examine if researcher-initiated reminders to pharmacists for follow-up can be replaced by more practical, on-site pharmacist reminder systems. Future research should also explore the exciting possibilities that community pharmacists can make an even greater impact in antidepressant education and monitoring since they are highly accessible in the community and have the ability to significantly influence behavioral and cognitive outcomes through the use of a relatively simple monitoring tool, a single 90-minute training session, and only three telephone calls.

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